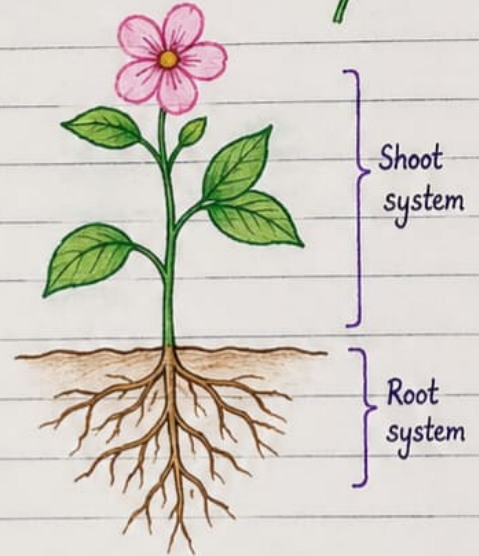


Morphology of Flowering Plants



Introduction

- ▶ Morphology is the study of external form and structure of plants.
- ▶ Flowering plants show great diversity in size, shape and habitat but their **basic body plan** is same.
- ▶ A typical flowering plant has two main parts:
 - 1) Root system (underground part)
 - 2) Shoot system (above ground part)



1. Root System

- ▶ Roots are generally **non-green**, underground parts.
- ▶ They anchor the plant in the soil.
- ▶ They absorb **water** and **minerals**.
- ▶ They may **store food** in some plants.

Types of Root Systems

(i) Tap Root System

- A well developed primary root is present.
- It gives lateral branches.
- Example → Pea, Mustard, Hibiscus



(ii) Fibrous Root System

- Primary root is absent or short lived.
- Many thin roots arise from the base of stem.
- Example → Wheat, Rice, Maize, Grass



Modifications of Roots

- Storage : Carrot, Radish, Sweet potato
- Support : Stilt roots in Maize, Banyan
- Climbing : In money plant
- Respiration : Pneumatophores in mangroves
- Parasitic : In Cuscuta

2. Shoot System

- ▶ The part of plant **above the ground** is called shoot system.
- ▶ It consists of **stem, leaves, buds, flowers** and **fruits**.
- ▶ **Functions** : Photosynthesis, support, reproduction, conduction.

Parts of Shoot System

- Stem
- Leaves
- Buds
- Flowers
- Fruits



★ Understanding the morphology helps in identifying, classifying and studying the functions of plants.

3. Root System

► Roots are of two types :

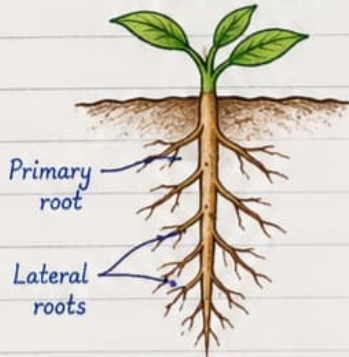
- **Tap root system** – One primary root grows vertically downward.
- **Fibrous root system** – Many roots of similar size arise from the base of stem.

Examples

- Tap root system : Pea, Mustard, Hibiscus, Sunflower
- Fibrous root system : Wheat, Rice, Maize, Onion

Types of Root Systems

(i) Tap Root System



One main root grows deep into the soil and gives out lateral roots.

Example – Pea

(ii) Fibrous Root System

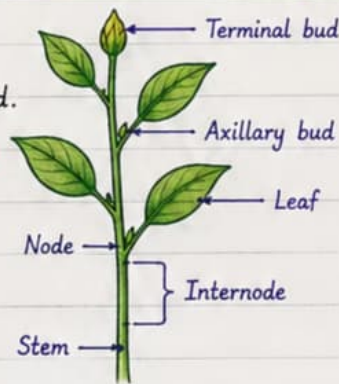


Many roots of equal size arise from the base of stem.

Example – Wheat

4. Shoot System

- Shoot system develops **above** the ground.
- It is made up of **stem, leaves, buds, flowers** and **fruits**.
- The stem bears **leaves** at **nodes**.
- The portion between two nodes is called **internode**.



Functions of Shoot System

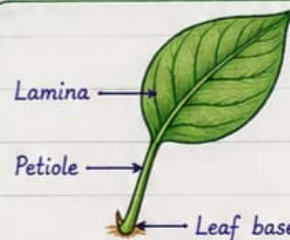
- **Photosynthesis** (by leaves)
- **Support**
- **Transport of water, minerals and food**
- **Reproduction** (flowers and fruits)
- **Storage** (in some stems)

Modifications of Shoot

Modification	Example	Function
Runner 	Grass, Strawberry	Helps in spreading and vegetative propagation
Rhizome 	Ginger, Turmeric	Storage of food and vegetative propagation
Tuber 	Potato	Storage of food
Bulb 	Onion, Garlic	Storage of food
Thorn 	Bougainvillea, Citrus	Protection

5. Leaves

- Leaves are lateral appendages which arise from nodes.
- They are the main sites of photosynthesis.
- A typical leaf has three parts –
(i) Leaf base (ii) Petiole (iii) Lamina



Venation – Arrangement of veins and veinlets in the lamina.

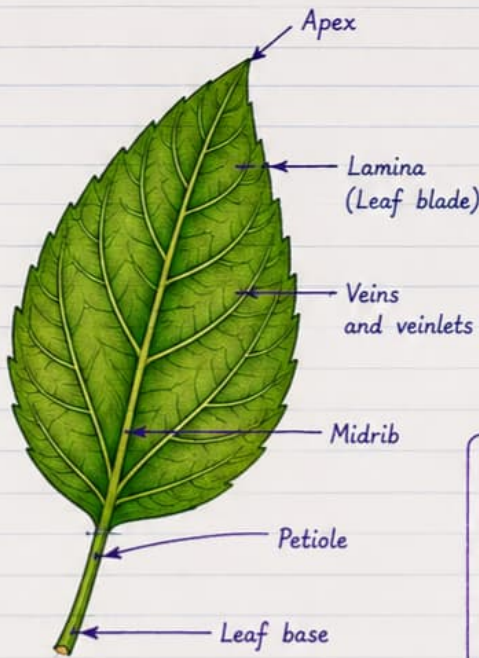
Types of Venation :

- (i) Reticulate (e.g., Mango)
- (ii) Parallel (e.g., Banana)

★ **Remember** : Root system anchors the plant and absorbs water & minerals, while shoot system helps in photosynthesis, transport and reproduction.

6. Leaf

- Leaves are the chief organs of **photosynthesis**.
- They are **flat, green** and **thin**.
- A typical leaf shows the following parts -
 - Leaf base** - Attached to the stem.
 - Petiole** - Stalk that connects leaf blade to the stem.
 - Lamina (Leaf blade)** - Broad, flat and green part.
 - Midrib** - Main vein running through the center.
 - Veins and veinlets** - Form a network for transport.



Leaf Venation

- Reticulate venation** - e.g., Mango, Hibiscus
- Parallel venation** - e.g., Banana, Grass

Modifications of Leaves

- Tendrils** - Pea
- Spines** - Cactus
- Pitchers** - Nepenthes
- Scales** - Onion
- Thorns** - Bougainvillea

Key Point : Leaf is the main site for photosynthesis and also helps in transpiration.

7. Inflorescence

- Inflorescence is the arrangement of flowers on the floral axis.
- It helps in better display of flowers and attracts pollinators.

Types of Inflorescence

(A) Racemose (Indeterminate)

Main axis continues to grow and does not end in a flower.

(B) Cymose (Determinate)

Main axis ends in a flower. Growth is limited.

Raceme



Flowers pedicellate.

Spike



Flowers sessile.

Catkin



Long, slender, drooping spike.

Corymb



Lower flowers pedicellate, upper flowers sessile.

Umbel



Pedicels arise from the same point.

Dichasium



Two lateral branches below the terminal flower.

Monochasium



One lateral branch below the terminal flower.

Helicoid cyme

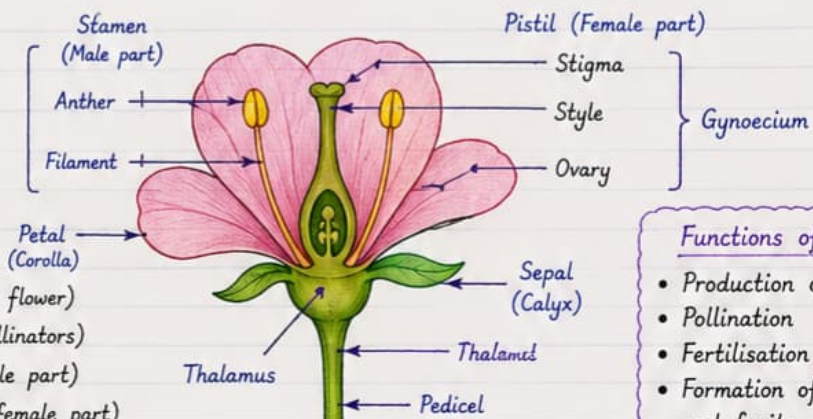


Lateral branches arise on the same side.

8. Flower

- Flower is the **reproductive** part of the plant.
- A typical flower is borne on a pedicel (stalk).
- It has four whorls attached to the thalamus.

- Calyx** - Sepals (protects the flower)
- Corolla** - Petals (attract pollinators)
- Androecium** - Stamens (male part)
- Gynoecium** - Pistil / Carpel (female part)



Functions of Flower

- Production of gametes
- Pollination
- Fertilisation
- Formation of seeds and fruits

Note : Morphology helps in **identification, classification** and understanding the **adaptations** of flowering plants.

9. Fruit

- ▶ Fruit is the **mature ovary**.
- ▶ It develops after fertilisation and contains **seeds**.
- ▶ Fruits **protect** the seeds and help in their **dispersal**.
- ▶ Fruits are of two types –
 1. **Simple fruits** – Develop from a single ovary of a flower.
 2. **Aggregate fruits** – Develop from many ovaries of a flower.
 3. **Multiple fruits** – Develop from the whole inflorescence.

Types of Fruits

(i) Simple Fruits

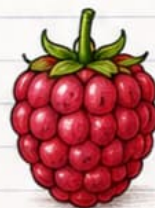
Develop from a single ovary of a flower.



Example – Mango

(ii) Aggregate Fruits

Develop from many ovaries of a single flower.



Example – Raspberry

(iii) Multiple Fruits

Develop from the whole inflorescence.



Example – Pineapple

10. Seed



- ▶ Seed contains an **embryo** and **food reserve**.
- ▶ It develops from the **ovule** after fertilisation.

Adaptations of Fruits and Seeds for Dispersal

By Wind

Light weight, small, with wings or hairs.

Example – Maple, Dandelion



By Water

Fruits are light and spongy.

Example – Coconut



By Animals

Fleshy fruits are eaten by animals. Seeds pass in their droppings.

Example – Guava, Cherry



By Explosion

Fruits burst open suddenly and throw out the seeds.

Example – Pea, Balsam



By Adhesion

Fruits or seeds have hooks or spines.

Example – Xanthium, Urena



Key Points

- ★ **Morphology** = Study of external form and structure.
- ★ **Root system** → Absorption, anchorage, storage.
- ★ **Shoot system** → Support, photosynthesis, transport, reproduction.
- ★ **Leaves** → Main site of photosynthesis.
- ★ **Flowers** → Reproductive organs.
- ★ **Fruits & seeds** → Protect seeds and help in dispersal.

Root System

- Tap root
- Fibrous root
- Functions – Anchorage, Absorption, Storage

Leaf

- Parts – Petiole, Lamina, Venation
- Functions – Photosynthesis, Transpiration

Quick Summary

Shoot System

- Stem, Leaves, Buds, Flowers, Fruits
- Functions – Support, Transport, Reproduction, Photosynthesis

Flower

- Parts – Calyx, Corolla, Androecium, Gynoecium
- Functions – Pollination, Fertilisation

Fruit & Seed

- Types of fruits – Simple, Aggregate, Multiple
- Seed – Contains embryo and food reserve

Remember : Understanding the morphology of flowering plants helps in better identification, classification and understanding their life processes.